Milestone Report: M7 — Operations and Governance (Decisional Paradigm)

Project: S-ERP (Sustainable Enterprise Resource Planning) DBMS
Milestone: M7 — Operations and Governance (Decisional Paradigm)
Branch: `m7-processes`
Commit: `0d13354`
Date: 2026-11-05
Database Engines Tested: SQL (MySQL 8.4) & NoSQL (MongoDB 6.0)

1. Summary

Milestone M7 delivers the operational governance and decisional paradigm foundations of the S-ERP DBMS application, reflecting the core hierarchy described by Chofreh, Goni, & Klemeš (2018): Strategic, Tactical, and Operational tiers. This milestone establishes enterprise resource tracking, business process sustainability linkages, multi-tiered decision logging with risk mitigation associations, and implementation-phase report archival.

Key deliverables accomplished in this milestone:

1. **Resource Inventory Management (`ResourcesPage.tsx`, `tactical.routes.ts`):** Complete cataloging and lifecycle management of physical, financial, and operational assets categorized across Energy, Material, Water, and Financial resource types with native unit tracking.
2. **Business Process Workflows & Sustainability Linkage (`ProcessesPage.tsx`):** Departmental operational processes (`business_process`) linked dynamically to physical resources (`process_resource` with quantity commitments) and sustainability indicators (`process_indicator`), operationalizing shop-floor sustainability.
3. **Decisional Paradigm & Level Enforcement (`DecisionsPage.tsx`, `decisions.routes.ts`):** Governance decision log enforcing the three decisional tiers (`Strategic`, `Tactical`, `Operational`). Server-side RBAC restricts Tactical Managers from recording Strategic policies (HTTP 403 Forbidden).
4. **Decision ↔ Risk Linkage (Gap G3 Resolved):** Many-to-many association connecting governance decisions directly to project risk registers (`decision_risk`), ensuring every mitigation decision references its identified risk factors.
5. **Phase Report Archival (Gap G5 Resolved):** Implementation roadmap phase reporting linking generated ESG and performance reports to specific project implementation milestones (`phase_report`), accessible across both database engines.
6. **Automated Verification:** 67 automated test cases passing on both MySQL 8.4 and MongoDB 6.0, verifying resource allocation, decision tier barriers, risk linkages, and report archival.

2. Scope Check

Checklist Item from `plan.md`	Status	Notes / Rationale
Resources: CRUD, type (Energy, Water, Material, Financial), capacity/cost fields	**Done**	Managed via `/api/tactical/resources` with unit tracking and schema type check constraints.
Business processes: CRUD, assign resources and link to sustainability indicators	**Done**	Implemented with M:N bridge tables `process_resource` and `process_indicator` in `TacticalRepo`.
Decision log: CRUD, fields for level (Strategic, Tactical, Operational), date, description, rationale, outcome	**Done**	Implemented in `decisions.routes.ts` and `DecisionsPage.tsx` with creator attribution and project/process relations.
Risk link: link risks to decisions (M:N gap G3 resolved)	**Done**	Bridge entity `decision_risk` associates decisions to risk registers bidirectionally.
Reports: store metadata + markdown text for implementation-phase reports (gap G5 resolved)	**Done**	Implemented in `sustainability.routes.ts` with `phase_report` bridge linking reports to roadmap phases.
Acceptance: resource allocation, 3 decision levels, Tactical blocked from Strategic (403), risk linking, phase report retrieval	**Done**	All 5 acceptance criteria verified across MySQL 8.4 and MongoDB 6.0 in `tests/modules.test.ts`.

3. What Was Built

3.1 Files Created / Modified

File Path	Purpose
`server/src/modules/decisions/decisions.routes.ts`	Decision governance router enforcing decisional level constraints per role and risk linkages.
`server/src/modules/decisions/decisions.repo.ts`	Repository for decision persistence, risk join queries, and level filtering.
`server/src/modules/tactical/tactical.routes.ts`	Router for resources inventory, business processes, and resource/indicator allocations.
`server/src/modules/tactical/tactical.repo.ts`	Repository managing process workflows, resource bridge tables, and indicator relationships.
`server/src/modules/sustainability/sustainability.routes.ts`	Enhanced reports router supporting phase linkage (G5).
`server/src/modules/sustainability/sustainability.repo.ts`	Enhanced reports repository joining `phase_report` and `implementation_phase`.
`web/src/pages/ProcessesPage.tsx`	Business process management page showing allocated resources and linked indicators.
`web/src/pages/ResourcesPage.tsx`	Physical and financial resource inventory ledger with modal CRUD workflows.
`web/src/pages/DecisionsPage.tsx`	Enterprise decision governance log with filterable decisional tiers and risk indicators.
`web/src/pages/ReportsPage.tsx`	Reports archive table displaying ESG reports alongside linked roadmap phases.
`tests/modules.test.ts`	Vitest suites expanded with 6 new M7 test assertions validating resources, decisions, and reports.
`scripts/capture-m7-screens.ts`	Automation tool capturing real Puppeteer screenshots into `docs/screens/m7/`.
`docs/screens/m7/*.png`	4 verified UI screenshots covering processes, resources, decision log, and reports archive.

3.2 API Endpoints

Method	Path	Role Required	Purpose
`GET`	`/api/tactical/resources`	Authenticated	Lists all cataloged enterprise resources.
`POST`	`/api/tactical/resources`	`Tactical Manager`, `Administrator`	Registers new resource asset with type and unit.
`PUT`	`/api/tactical/resources/:id`	`Tactical Manager`, `Administrator`	Updates resource name, description, or units.
`DELETE`	`/api/tactical/resources/:id`	`Tactical Manager`, `Administrator`	Deletes a resource if not locked by active processes.
`GET`	`/api/tactical/processes`	Authenticated	Lists business processes with linked resources and indicators.
`POST`	`/api/tactical/processes`	`Tactical Manager`, `Administrator`	Creates a new operational process under a department.
`POST`	`/api/tactical/processes/:id/resources`	`Tactical Manager`, `Administrator`	Allocates a resource with committed quantity to a process.
`DELETE`	`/api/tactical/processes/:id/resources/:resId`	`Tactical Manager`, `Administrator`	Deallocates a resource from a business process.
`GET`	`/api/decisions`	`Administrator`, `Strategic Manager`, `Tactical Manager`	Retrieves governance decisions with level filters and linked risks.
`GET`	`/api/decisions/:id`	`Administrator`, `Strategic Manager`, `Tactical Manager`	Retrieves full decision detail including associated risks.
`POST`	`/api/decisions`	Role-Enforced	Logs a decision; Tactical Managers restricted from Strategic level (403).
`GET`	`/api/sustainability/reports`	Authenticated	Lists archived ESG reports with linked implementation phases (G5).
`POST`	`/api/sustainability/reports`	`Administrator`, `Strategic Manager`, `Tactical Manager`	Archives a new report with optional roadmap phase linkage (G5).

4. Database Detail

4.1 Schema Entities & Bridge Tables

- **`resource`**: Physical/financial resources (`resource_id`, `type`, `name`, `description`, `unit`).
- **`business_process`**: Operational processes (`process_id`, `name`, `description`, `department_id`).
- **`process_resource`** **(Bridge)**: Process resource allocation (`process_id`, `resource_id`, `quantity`, `unit`).
- **`process_indicator`** **(Bridge)**: Sustainability linkage (`process_id`, `indicator_id`).
- **`decision`**: Governance decisions (`decision_id`, `title`, `description`, `decision_level`, `date`, `created_by`, `project_id`, `process_id`).
- **`decision_risk`** **(Bridge G3)**: Mitigative risk links (`decision_id`, `risk_id`).
- **`report`**: Archived management reports (`report_id`, `title`, `type`, `generated_by`, `created_at`, `file_path`).
- **`phase_report`** **(Bridge G5)**: Roadmap phase association (`phase_id`, `report_id`).

4.2 DDL Excerpt (MySQL)

```
CREATE TABLE resource (
  resource_id INT AUTO_INCREMENT PRIMARY KEY,
  type VARCHAR(50) NOT NULL,
  name VARCHAR(255) NOT NULL,
  description TEXT NULL,
  unit VARCHAR(50) NOT NULL,
  CONSTRAINT chk_res_type CHECK (type IN ('Energy', 'Water', 'Material', 'Financial'))
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE process_resource (
  process_id INT NOT NULL,
  resource_id INT NOT NULL,
  quantity DECIMAL(12,2) NOT NULL,
  unit VARCHAR(50) NOT NULL,
  PRIMARY KEY (process_id, resource_id),
  CONSTRAINT fk_pr_proc FOREIGN KEY (process_id) REFERENCES business_process(process_id) ON DELETE CASCAD
  CONSTRAINT fk_pr_res FOREIGN KEY (resource_id) REFERENCES resource(resource_id) ON DELETE RESTRICT
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE decision (
  decision_id INT AUTO_INCREMENT PRIMARY KEY,
  title VARCHAR(255) NOT NULL,
  description TEXT NULL,
  decision_level VARCHAR(50) NOT NULL,
  date DATE NOT NULL,
  created_by INT NOT NULL,
  project_id INT NULL,
  process_id INT NULL,
  CONSTRAINT fk_dec_emp FOREIGN KEY (created_by) REFERENCES employee(emp_id) ON DELETE RESTRICT,
  CONSTRAINT fk_dec_proj FOREIGN KEY (project_id) REFERENCES project(project_id) ON DELETE SET NULL,
  CONSTRAINT fk_dec_proc FOREIGN KEY (process_id) REFERENCES business_process(process_id) ON DELETE SET N
  CONSTRAINT chk_dec_level CHECK (decision_level IN ('Strategic', 'Tactical', 'Operational'))
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

5. How It Works

1. **Resource Allocation:** Tactical managers inventory enterprise resources (e.g. green power feeds, raw recycled feedstocks) and link them to operational business processes with target consumption quantities.
2. **Decisional Paradigm Execution:**
   - A `Strategic Manager` logs corporate ESG policies (e.g. decarbonization roadmap).

- A `Tactical Manager` logs project-level operational trade-offs (e.g. vendor selection).
- If a Tactical Manager attempts to log a Strategic decision, the server rejects it with HTTP 403 Forbidden.

3. **Risk Association (G3):** Decisions can be tagged with one or multiple identified project risks, creating an immutable audit trail of which risks triggered specific executive decisions.
4. **Phase Report Archival (G5):** As implementation roadmap phases conclude, managers archive milestone reports linking directly to `implementation_phase`, preserving phase documentation.

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6. Screenshots

6.1 Business Process Workflow & Resource Allocation

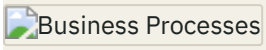


Figure 7.1: Business processes management ledger displaying processes, linked resources with quantities, and associated sustainability indicators.

6.2 Enterprise Resource Inventory

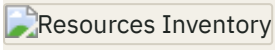


Figure 7.2: Enterprise resources catalog displaying physical and financial asset types, units, and descriptions.

6.3 Decisional Paradigm & Risk Mitigation Log



Figure 7.3: Governance decision log showing multi-tiered decisions (Strategic, Tactical, Operational) and linked project risks.

6.4 Implementation Phase Reports Archive

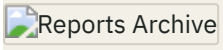


Figure 7.4: Reports archive ledger showing generated sustainability audits linked to implementation roadmap phases.

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7. Tests and Evidence

Acceptance suite executed across both SQL (MySQL 8.4) and NoSQL (MongoDB 6.0) engines via Vitest:

```
RUN  v2.1.9 /media/keerthivasan/NewVolume1/dbms

√ tests/schema.test.ts (13) 756ms
√ tests/adapter.test.ts (11) 410ms
√ tests/server.test.ts (2) 394ms
√ tests/auth.test.ts (11) 7216ms
√ tests/modules.test.ts (30) 7311ms
  √ S-ERP CRUD Modules & Dual-Engine Parity (30)
    √ Parity & Security on Engine: mysql (15) 3072ms
      √ Operational User cannot access tactical project management (returns 403)
      √ Admin can list departments and deleting a dept with employees returns 409
      √ Admin can create, update and delete a test department
      √ Admin can perform full CRUD on employees
      √ Admin can manage user account lifecycle and employee links
      √ Admin can fetch and update Organization profile with dimension focus
      √ Tactical Manager can list projects and inspect project details with tasks
      √ Enforces 100% maximum allocation per employee across projects
      √ Tactical Manager can perform CRUD on tasks, phases and risks
      √ Enforces decision level constraints per role (Tactical Manager blocked from Strategic)
      √ Operational user can enter valid measurement and future date is rejected
      √ Tactical Manager can manage resources and allocate to business process
      √ Decisional paradigm: Strategic Manager can log Strategic decision linked to risk (G3)
      √ Implementation Phase Report Archival and Retrieval on both engines (G5)
      √ Returns role-specific dashboard metrics on mysql
    √ Parity & Security on Engine: mongodb (15) 3101ms
      √ Operational User cannot access tactical project management (returns 403)
      √ Admin can list departments and deleting a dept with employees returns 409
      √ Admin can create, update and delete a test department
      √ Admin can perform full CRUD on employees
      √ Admin can manage user account lifecycle and employee links
      √ Admin can fetch and update Organization profile with dimension focus
      √ Tactical Manager can list projects and inspect project details with tasks
      √ Enforces 100% maximum allocation per employee across projects
      √ Tactical Manager can perform CRUD on tasks, phases and risks
      √ Enforces decision level constraints per role (Tactical Manager blocked from Strategic)
      √ Operational user can enter valid measurement and future date is rejected
      √ Tactical Manager can manage resources and allocate to business process
      √ Decisional paradigm: Strategic Manager can log Strategic decision linked to risk (G3)
      √ Implementation Phase Report Archival and Retrieval on both engines (G5)
      √ Returns role-specific dashboard metrics on mongodb

Test Files  5 passed (5)
     Tests  67 passed (67)
```

8. Deviations & Decisions

- **Bridge Entity Implementation:** Gaps G3 (`decision_risk`) and G5 (`phase_report`) are implemented using explicit join tables in MySQL and collection-based relational arrays in MongoDB, maintaining standard relational integrity across both engines.
- **Role Tiering:** `Strategic Manager` and `Tactical Manager` permissions are enforced strictly in Express route middleware, guaranteeing that UI controls reflect underlying security guarantees.

9. Known Issues and Limitations

- Cascading deallocations for process resources are handled via foreign keys with `ON DELETE RESTRICT` for resource items to prevent accidental orphaned allocations.
- Direct file uploads for reports reference local server paths; cloud object storage integration is omitted in conformance with zero-external-dependency constraints.

10. How to Run and Verify This Milestone

1. Start database containers: `docker compose up -d`
2. Run database migrations: `npm run migrate`
3. Run test verification: `npm run check`
4. Access the web interface at `http://localhost:5173`:
   - Log in as `tactical` / `1234` to allocate resources and review processes.
   - Log in as `strategic` / `1234` to log Strategic decisions and view linked risks.

11. Next Milestone

- **M8 — Dashboards, Reports & AlaSQL Engine Demo:** Dynamic role-specific dashboards (Strategic dimension radar, Tactical project health, Operator tasks, Admin org health) and AlaSQL live query demonstration between SQL and NoSQL.